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Pricing and Trading Strategies of Chinese Convertible Bonds

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Abstract

Convertible bond is a popular tool for financing, trading and risk management. In China, this kind of products bears complicated features that present challenges for pricing. In this thesis, we address the problem of pricing Chinese convertible bonds(CCB) by Monte Carlo simulation and dynamic programming.

At each exercising time, we use the state variables of the underlying stock to regress the continuation value, and then we apply standard backward induction to get the coefficients until the moment of time zero, thus the price of the CCB is obtained. We apply the pricing of CCBs by simulation and test the performance of an underpriced strategy: long the 10 most underpriced CCBs and rebalance daily. The result show this strategy significantly outperforms the double-low strategy which is used as a benchmark.

Moreover, we propose an enhanced rotational multi-factor quantitative model for CCB selection based on machine learning techniques. We select factors that accurately reflect market information by Information Coefficient (IC) analysis and a stratified backtesting approach, and construct a rotating multi-factor stock selection framework based on various machine learning model. To further validate the effectiveness of our approach, we implement a multi-factor CCB selection strategy based on SVM and linear regression. Through trading simulations spanning 118 trading days (from February to July 2023), we achieved annualized returns of 13.22 and 18.62 percentage respectively with a transaction cost of 0.1%.

The study contributes to the literature of convertible bond pricing and quantitative investments. We treat the downward adjustment clause as a probabilistic event when the put provision is triggered. In this way, we combine the downward adjustment clause with put provision in a simple manner.

Moreover, we conduct systematic empirical analysis of the factors in CCBs, identifying factors that truly have potential profit effects, and confirm the ability of machine learning-driven convertible bond factor strategy to obtain excess returns.

Key Words: Chinese convertible bonds; Pricing; Machine learning; Monte Carlo simulation; Downward adjustment.

摘要

可轉換債券是一種很受歡迎的融資、交易和風險管理產品。在中國, 此類產品具有较複雜的特點, 為定價帶來了许多挑戰。在本文中, 我們透過蒙特卡羅模擬和動態規劃來解決中國可轉換債券（CCB）的定價問題。

在每個行使時間，我們使用標的股票的狀態變數來回歸連續值, 然後應用標準後向歸納法得到直到時間為零的時刻的係數, 進而得到 CCB 的價格。我們透過模擬來進行 CCB 定價, 並測試定價策略的表現：先假設交易成本為 0.1%, 做多 10 個定價最低估的 CCB, 並在每日重新平衡。結果表明, 該策略明顯優於作為基準的雙低策略。

我們還提出了一種基於機器學習技術的 CCB 選擇的增強版多因子量化模型。我們透過信息係數（IC）分析和分層回測方法選擇準確反映市場信息的因子, 並建立基於各種機器學習模型的轉動多因子選股框架。為了進一步驗證我們方法的有效性, 我們實施了基於 SVM 和線性迴歸的多因子 CCB 選擇策略。同樣假設交易成本為 0.1%, 透過跨越 118 個交易日（2023 年 2 月至 2023 年 7 月）的交易模擬, 我們分別獲得了 13.22% 和 18.62% 的年化收益。

該研究為可轉換債券定價和量化投資做出了貢獻。我們將下修條款視為回售條款觸發時的機率事件。這樣, 我們就將下修條款和回售條款簡單地結合在一起。此外, 我們對 CCB 的因子進行了系統性的實證分析, 識別出真正具有潛在利潤效應的因子, 並證實了機器學習驅動的可轉債多因子策略獲得超額收益的能力。

關鍵字：中國可轉債；定價；機器學習；蒙特卡羅模擬；下修條款。

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Chapter 1

Introduction

1.1 Chinese Convertible Bonds

Chinese convertible bond market has been growing fast in recent years, attracting attention from both domestic and international investors.

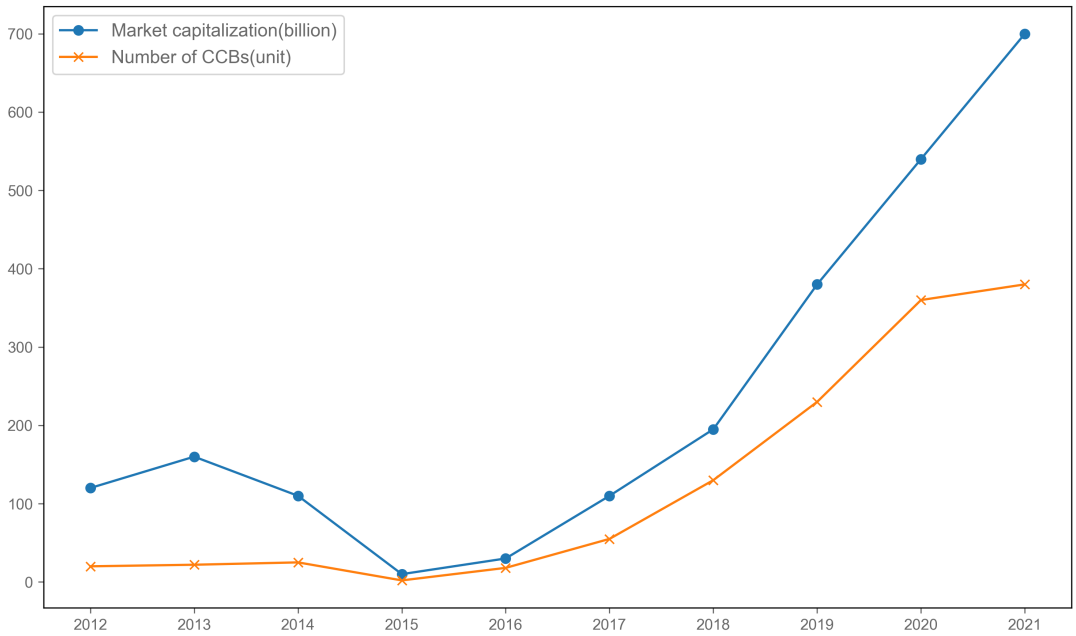


Figure 1.1: The growth of Chinese convertible bond market. (Data source: Wind)

A convertible bond is a type of fixed-income security that can be converted into the issuer's common stocks. Chinese convertible bonds are embedded with some other special clauses, such as call/put provision and downward adjustment.

Here are some typical features of CCBs.

- Bond: A CCB has face value and coupon rate.
- Conversion to stocks: CCBs can be converted into stocks at a certain ratio during the conversion period (typically 6 months after issuance).
- Call/put provision: The m-out-of-n call/put provision is, when the underlying stock price closes above/below the call/put triggered price for any m_c/m_p days over the past n_c/n_p consecutive trading days, the call/put provision is triggered. The call provision stipulate that the investors have to convert the CCB into stocks or sell it to issuers within a period of time. The put provision stipulate that the investors have the right to put back the CCB to issuers at the put price.
- Downward adjustment: Similarly to the call/put provision, when the underlying stock price is below a pre-set price for pre-set m days over the past n consecutive trading days, the issuers can lower the conversion price to make the conversion value higher and more attractive to investors.
- Redemption at maturity: On the CCB's maturity date, the investors can sell the CCB back to the issuers at a pre-set price.

Notations	Explanations
C_t	Conversion price at time t
S_t	Underlying stock price at time t
FV	Face value
m	Conversion ratio, $m = \frac{FV}{C_t}$
CV	Conversion value, $CV = mS_t$
σ	Stock volatility
BP	Market price
PR	Conversion premium rate, $PR = \frac{BP-CV}{CV}$
B	Redemption price at maturity

Table 1.1: Some key notations of CCBs.

1.1.1 An Example of CCB

Here is the basic information of the Daqin convertible bond from Jisilu.

Price	120.480
Conversion value	121.22
Conversion premium rate	-0.61%
Industry	railway transport
Start date	2020-12-14
Listing date	2021-01-15
Maturity date	2026-12-14
Time to maturity	2.778
Conversion start date	2021-06-18
Put option start date	2024-12-13
Conversion price	6.22
Put price	100.00+ accumulated unpaid interest
Redemption price at maturity	108.00
Issuer rating	AAA
Bond rating	AAA
Call triggered price	130% of the conversion price
Put triggered price	70% of the conversion price
Downward triggered price	85% of the conversion price
m_c/n_c	15/30
m_p/n_p	30/30
m/n	15/30

Table 1.2: Overview of Daqin CB.

1.2 Literature Review

1.2.1 Convertible Bond Pricing

The valuation models for convertible bonds can be mainly classified into two types [杨立洪 et al., 2010]: separation approach and aggregate approach.

- Separation approach: In this method, the convertible bond is divided into a combination of simple assets, such as "bond + option". The advantage is that it's simple and easy to understand. However, it may not capture the complex features of CCBs because CCBs include various clauses : conversion , call/put provision, downward adjustment. Conversion and put option are held by in-

vestors, while downward adjustment and call option are held by the issuers. There is a complex game between investors and issuers to exercise their options, which leads to the difficulty of pricing a CCB [Ingersoll Jr, 1977].

- Aggregate approach: We treat a CCB as a "black box", and can observe the cash flows generated by the CCB's clauses in simulation. Valuation is performed by discounting the expected cash flows. Well-known numerical methods used in this approach include Monte Carlo simulation [Ammann et al., 2008], binomial tree [Milanov and Kounchev, 2012], finite difference method [Brennan and Schwartz, 1978], etc. These numerical methods can handle more complex bond provisions, but have low computational efficiency.

Monte Carlo simulation is quite suitable for modeling discrete coupon and dividend payments, capturing more realistic dynamics of underlying state variables, and incorporating path-dependent call features. Path dependencies, such as conditions for early redemption based on stock price thresholds over a specific number of days, can be effectively accounted for. It is efficient to deal with multiple state variables and flexible to expand.

In recent years, a considerable number of important articles have addressed the problem of pricing American-style options by using a combination of Monte Carlo simulation and dynamic programming, such as [Longstaff and Schwartz, 2001]. Their central point lies in proposing a Least Squares framework to regress the continuation value of American-style options. [Lvov et al., 2004] and [Crépey and Rahal, 2011] extended the Least Squares method of [Longstaff and Schwartz, 2001] and [Carriére, 1996] to price game options. Our thesis embraces this method and extends it to CCBs.

The CCB market started to develop gradually after the year 2000. With the growth of CCB market, [杨立洪 et al., 2010] analyzed the pricing problem of CCBs using a binary tree method with a sample of 24 CCBs. However, previous studies have used various models to theoretically price convertible bonds, none of them considered

the impact of downward adjustment clauses on pricing until [刘大巍 et al., 2011]. They used the Least Squares Monte Carlo method to analyze certain convertible bond pricing issues, and suggested considering downward adjustment clauses to improve the accuracy of CCB pricing.

1.2.2 Trading Strategies of Convertible Bonds

The application of machine learning in the field of financial investment has attracted widespread attention in recent years. Multiple studies have shown that machine learning methods can be successfully used for stock return prediction and fundamental analysis. For example, research by [Cao and You, 2024] found that using historical data of stocks can help fundamental analysts make better investment decisions, especially for models that can handle nonlinear relationships. These models have generated more accurate and informative predictions compared to a series of state-of-the-art earnings forecast models. [Gu et al., 2020] conducted an empirical study on asset pricing using machine learning methods, and their research shows that machine learning models can be used to predict stock prices and returns, demonstrating its potential in improving trading strategies.

Other studies were focused on multi-factor trading strategies for convertible bonds. [开源证券, 2021] and [海通证券, 2022] have conducted preliminary investigations into multi-factor strategies for convertible bonds. These studies use different factors and models to analyze the premium rates and investment strategies of convertible bonds. Overall, the researches on factors are mainly conducted through grouping and analysis of differences and multi-factor strategies are mainly constructed through linear regression methods.

[Tolstikhin et al., 2021] applies multi-layer perceptron (MLP) architecture to computer vision and [Kavzoglu and Teke, 2022] uses NGBoost and XGBoost to delineate potential landslide-prone areas. People are very interested in using machine learning

to implement multi-factor strategies. [Li et al., 2019] applied 12 machine learning algorithms to analyze 96 anomaly factors in the A-share market. The results show that machine learning algorithms can effectively identify complex patterns between factors and excess returns, and its investment strategy can achieve better performance than traditional linear algorithms and all single factors.

There are two main methods to implement multi-factor strategies:

- Multiple linear regression method:

$$R = \alpha + \theta_1 f_1 + \theta_2 f_2 + \theta_3 f_3 + \ldots\ldots \theta_n f_n + \epsilon,$$

  where R is the rate of return, α is the intercept, f_i is the value of factor i , θ_i is the coefficient corresponding to the f_i and ϵ is the error.

  This method assume that the factors are independent variables and the relationship between factors and stock returns is linear, and other parts are all unobservable errors. The Fama-French three-factor model is based on this method.

  In fact, people always set the coefficients of each factor to the same value, as seen in [开源证券, 2021] , [Li et al., 2019] and [O'Shaughnessy, 1998], it is called simple equal-weight method.

- Machine learning regression method:

$$R = g(f; \theta) + \epsilon,$$

  where $f = (f_1, \cdots, f_n)$, $\theta = (\theta_1, \cdots, \theta_n)$ and g is a function of abstract representation, corresponding to different machine learning regression methods, and other notations are the same as before. According to [Gu et al., 2020], machine learning is particularly appropriate for problems with uncertain func-

tional forms and it is explicitly designed to capture intricate nonlinear relationships. Additionally, applying penalties to parameters and employing stringent model selection criteria help mitigate overfitting and prevent false discoveries.

[Gu et al., 2020] found that decision trees and neural networks are the best-performing machine learning methods, these two methods can capture the nonlinear interaction effects between variables that other methods tend to overlook. These nonlinear interactions play a significant role in asset pricing, which traditional regression methods often fail to capture due to their inability to handle such complexity [Li et al., 2019]. This thesis adopts machine learning regression algorithms to predict CCBs' price and build multi-factor trading strategies.

1.3 Thesis Structure

The remaining of the thesis is organized as follows:

In Chapter 2, we will introduce CCB valuation model using Least Squares Monte Carlo simulations.

Chapter 3 provides a description of the data set and examines the performance of the pricing algorithm by comparing the actual prices with the model prices of 10 CCBs with 3A credit rating in the first half of 2023, then we use the predicted price as a determining factor for backtesting in the first half of 2023 and observe the rate of return.

In Chapter 4, we select factors that accurately reflect market information through IC analysis and stratified backtesting method. Then we propose a multi-factor CCB strategy to achieve 18% annualized return.

Chapter 5 presents the conclusion of this thesis.

Chapter 2

Monte Carlo Simulations for CCB Pricing

In this chapter, we extend the Least Squares method of [Longstaff and Schwartz, 2001] and [Beveridge and Joshi, 2011] to price CCBs. Downward adjustment presents a challenging aspect within the CCB pricing problem, we treat the triggering of downward adjustment as a probabilistic event when put option is triggerd, thus it is compatible with the Least Squares pricing framework. In addition, we prove that there is a unique solution using Least Squares method to regress the continuation value at each exercise time and implement the "Multiple Regression" in [Beveridge and Joshi, 2011] on CCBs.

2.1 The Problem Setup

The valuation of CCB mainly depends on the evolution of the underlying stock price and the dynamic game between the investor and the issuer. We assume the underlying stock price S_t follow a risk-neutral dynamics as geometric Brownian motion with a constant σ and r:

$$dS_t = (r - q)S_t dt + \sigma S_t dW_t,$$

where r is the risk-free rate and q denotes the dividend yield. The time unit is per day.

We can obtain the solution by Ito's lemma as follows:

$$S_t = S_{t-1} e^{(r-q-\sigma^2)+\sigma(W_t - W_{t-1})}.$$

We represent the value of the convertible bond at time t as $V(t)$, and refer to the notation of game options of [Beveridge and Joshi, 2011].

$$\begin{aligned} V(t) &= \sup_{\tau_A \in \Gamma_t^A} \inf_{\tau_B \in \Gamma_t^B} \mathbb{E}\left[f_t\left(\tau_A, \tau_B\right) \mid \mathcal{F}_t\right], \\ &= \inf_{\tau_B \in \Gamma_t^B} \sup_{\tau_A \in \Gamma_t^A} \mathbb{E}\left[f_t\left(\tau_A, \tau_B\right) \mid \mathcal{F}_t\right]. \end{aligned}$$

Here are some extra notations.

- $\Xi^A = \{T_1, \ldots, T_m\}$: the set of exercise times for A, where

$$0 = T_0 < T_1 < \ldots < T_m = T.$$

- $\Xi^B = \{T_1, \ldots, T_m\}$: the set of exercise times for B, where

$$0 = T_0 < T_1 < \ldots < T_m = T.$$

- Γ_t^A : the set of possible exercise strategies for A taking values in $[t, T] \cap \Xi^A$.
- Γ_t^B : the set of possible exercise strategies for B taking values in $[t, T] \cap \Xi^B$.
- $f_t(\tau_A, \tau_B)$: the deflated cash-flows received by A on $[t, T]$.

- $V(t)$: the value of the convertible bond at time t.

"Deflating" in [Beveridge and Joshi, 2011] is a process of obtaining cash flow through step-by-step discounting at each exercise time from T to t, where each discounted cash is multiplied by the discount factor e^{-r} in one time interval, see [Beveridge and Joshi, 2011] and [Bielecki et al., 2008] for details.

Next we will show how to implement this abstract definition concretely.

2.2 Least Squares Regression

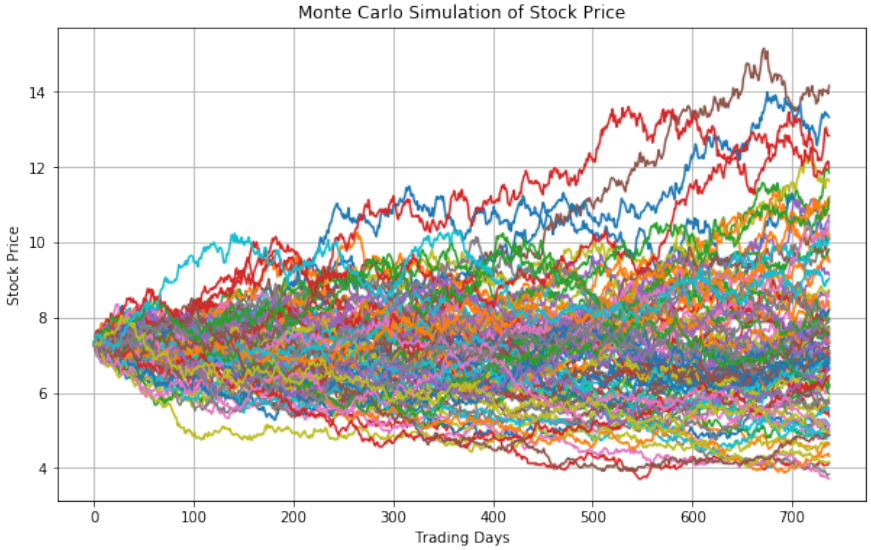


Figure 2.1: Simulation of stock price for Daqin CB.

Without loss of generality, we assume that A is the investor and B is the issuer. As we mentioned in Section 1, a CCB can have several possible states: forced redemption (call), putback (put), downward adjustment, conversion to stocks, continuation. This indicates that at each time , τ_A can select from four states: call, put, conversion, continuation. Due to the lack of information from B's perspective, this thesis only considers the maximum value of exercise from A's perspective.

We use F_t to represent the proportion of days when stock prices exceed the call triggered price k_t in the previous m_c days at time t (including time t). Similarly, we use Y_t to represent the proportion of days within the previous m_p days at time t when the stock price is lower than the put triggered price p_t. Denote p_F/p_Y is the threshold for F_t/Y_t to trigger call/option revision, in most cases, $p_F = 0.5$ and $p_Y = 1$.

Framework of regression

We combine the preliminary evaluation mentioned above with the Least Squares method in [Beveridge and Joshi, 2011] to propose the following framework:

1. The duration of the convertible bonds is T days, and they can only be exercised at the specified conversion period. We apply Monte Carlo simulation to generate M paths of the underlying stock price over T days.

   For path $i = 1, 2, \cdots, M$

   $$\begin{cases} S_t^{(i)} = S_{t-1}^{(i)} e^{(r-q-\sigma^2)+\sigma Z_t^{(i)}}, & t = 1, 2, \cdots, T, Z_t^{(i)} \sim N(0,1), \\ S_0^{(i)} = S_0. \end{cases}$$

   and then we do simulation on the M paths to price a specific CCB from $t = T$ to $t = 0$ and call it "backtrack".

2. Denote $V_t^{(i)}$ as the maximum cash flow back to t. First, at maturity T, the exercise value is the maximum value of conversion and redemption at maturity,

   $$V_T^{(i)} = \max(mS_T^{(i)}, B).$$

3. At conversion time t $(< T)$, we take the continuation value at ith path as

   $$y_t^{(i)} = e^{-r} V_{t+1}^{(i)},$$

and then we use bases $\mathbf{e_t} = \{S_t, {S_t}^2, F_t, {F_t}^2, Y_t, {Y_t}^2, S_tF_t, S_tY_t, F_tY_t\}$ to regress the continuation value using Least Squares and we can get regression's continuation value $\hat{y}_t^{(i)}$ for each path.

Regression at time t.

Path	$\mathbf{e_t}$	y_t
1	${\mathbf{e_t}}^{(1)}$	$y_t^{(1)}$
2	${\mathbf{e_t}}^{(2)}$	$y_t^{(2)}$
$\cdots$	$\cdots$	$\cdots$
M	${\mathbf{e_t}}^{(M)}$	$y_t^{(M)}$

Consider the optimal decision for each path, i.e., compare the values of $mS_t^{(i)}$, $\hat{y}_t^{(i)}$, K_t(if $F_t \geq p_F$) and P_t(if $Y_t \geq p_Y$) and choose the strategy with the maximum potential profit, which could be conversion, call, put or continuation, and thus we can determine whether to execise or continue.

$$V_t^{(i)} = \begin{cases} \max\{mS_t^{(i)}, K_t\} & \text{if } F_t \geq p_F, \\ \max\{mS_t^{(i)}, \hat{y}_t^{(i)}, P_t \cdot \mathbb{I}\{Y_t \geq p_Y\}\} & \text{if } F_t < p_F. \end{cases}$$

4. The convertible bond is priced by discounted cash flows back to time $t = 0$ using the risk-free interest rate and averaging over all paths,

$$V_0 = \frac{1}{M} \sum_{i=1}^{M} V_0^{(i)}.$$

5. Downward adjustment: when $Y_t = 1$, we consider the probability p of a downward adjustment occurring.

   To determine the value of p, we should identify the probability of downward adjustment upon putback provision for the CCB in the corresponding industry.

However, due to data limitations and considering that downward adjustment upon putback provision are commonly observed in practice, we set the parameter p to a relatively high value (such as 0.8).

Taking into account the uncertainty of the magnitude of downward adjustment and the fact that CCB downward adjustment is usually to the bottom-line, we assume the issuer revises to the minimum, i.e., C_t revises to the maximum of the average underlying stock prices over the previous 20 trading days and the last trading day. Let $\hat{C}_t$ be the adjusted conversion price, and $\hat{m}$ be the corresponding conversion ratio,

$$V_t^{(i)} = \max\{\hat{m}S_t^{(i)}, \hat{y}_t^{(i)}, P_t\}.$$

The existence and uniqueness of the solution for Least Squares regression at each time t (point 3 on page 12).

First, we model the continuation value at time t as:

$$y_t \approx \mathbf{e}_t\theta_t,$$

where

$$\theta_t = (\lambda_{1,t}, \lambda_{2,t}, \cdots, \lambda_{N,t})^{\mathrm{T}}, \mathbf{e}_t = (e_{1,t}, e_{2,t}, \cdots, e_{N,t}).$$

To approximate the model using Least Squares, we aim to find the coefficients θ_t that minimize the squared difference between the estimated continuation value $\hat{y}_t$ and the actual discounted continuation value:

$$\arg\min_{\theta_t} \sum_{i=1}^{M} \left(y_t^{(i)} - \mathbf{e}_t^{(i)}\theta_t\right)^2,$$

where $\{y_t^{(i)}\}_{1\leq i\leq M}$ is the actual discounted continuation value of M paths.

The optimal problem can be written as:

$$\arg\min_{\theta_\mathbf{t}} \|\mathbf{y_t} - \mathbf{E_t}\theta_\mathbf{t}\|_2^2 ,$$

where

$$\mathbf{y_t} = (y_t^{(1)} \ldots, y_t^{(M)})^\mathrm{T} \quad \text{and} \quad \mathbf{E_t} = (\mathbf{e}_t^{(1)}, \ldots, \mathbf{e}_t^{(M)})^\mathrm{T}.$$

Let

$$\mathcal{L}(\theta_\mathbf{t}) = \|\mathbf{E_t}\theta_\mathbf{t} - \mathbf{y_t}\|_2^2 ,$$

then

$$\mathcal{L}(\theta_\mathbf{t}) = (\mathbf{E_t}\theta_\mathbf{t} - \mathbf{y_t})^\mathrm{T}(\mathbf{E_t}\theta_\mathbf{t} - \mathbf{y_t}) = \theta_\mathbf{t}{}^\mathrm{T}\mathbf{E_t}{}^\mathrm{T}\mathbf{E_t}\theta_\mathbf{t} - 2\mathbf{y_t}{}^\mathrm{T}\mathbf{E_t}\theta_\mathbf{t} + \mathbf{y_t}{}^\mathrm{T}\mathbf{y_t},$$

$$\nabla\mathcal{L}(\theta_t) = 2\mathbf{E_t}{}^\mathrm{T}(\mathbf{E_t}\theta_\mathbf{t} - \mathbf{y_t}).$$

The optimization problem

$$\arg\min_{\theta_\mathbf{t}} \mathcal{L}(\theta_\mathbf{t})$$

is a convex optimization problem, thus

$$\theta_\mathbf{t}^* = \arg\min_{\theta_\mathbf{t}} \mathcal{L}(\theta_\mathbf{t}) \Leftrightarrow \nabla\mathcal{L}(\theta_\mathbf{t}^*) = \mathbf{0}.$$

Setting the gradient to zero and due to convexity of $\mathcal{L}$, the optimal solution $\theta_\mathbf{t}^*$ satisfies

$$\mathbf{E_t}{}^\mathrm{T}\mathbf{E_t}\theta_\mathbf{t}^* = \mathbf{E_t}{}^\mathrm{T}\mathbf{y_t}.$$

$\mathbf{E_t}$ is an $M \times N$ matrix, where in practice, M is much larger than N(we set $N = 8$), therefore $\mathbf{E_t}$ can be treated as a column full-rank matrix, so $\mathbf{E_t}{}^\mathrm{T}\mathbf{E_t}$ is an invertible $N \times N$ matrix. Thus the Least Squares problem always has a unique solution:

$$\theta_\mathbf{t}^* = (\mathbf{E_t}{}^\mathrm{T}\mathbf{E_t})^{-1}\mathbf{E_t}{}^\mathrm{T}\mathbf{y_t}.$$

In general, the exercise strategy at time t can be summarized as following.

Payoff	Condition	Decision
K_t	$F_t \geq p_F$ and $K_t \geq mS_t$	forced redemption
mS_t	$F_t \geq p_F$ and $K_t < mS_t$	forced conversion
P_t	$Y_t \geq p_Y, P_t \geq y_t$ and $P_t \geq mS_t$	putback
y_t	$Y_t \geq p_Y, y_t > P_t$ and $y_t \geq mS_t$	continuation
mS_t	$Y_t \geq p_Y, mS_t > P_t$ and $mS_t > y_t$	conversion
mS_t	$y_t \leq mS_t$	voluntary conversion
y_t	$y_t > mS_t$	continuation
B	$mS_T \leq B$	redemption at maturity
mS_T	$mS_T > B$	conversion at maturity

Table 2.2: Exercise strategy at each exercise time t (point 2 and 3 on page 12).

Improvement for the Least Squares regression framework

According to [Beveridge and Joshi, 2011], the performance of convertible bonds depends on underlying stock price S_t. When stock prices are high and conversion is likely to occur, convertible bonds exhibit equity-like characteristics. Conversely, in situations of significantly depressed stock prices, convertible bonds tend to resemble corporate bonds more closely.

In the point 3 of the regression framework on page 12, we use the idea of "Multiple Regression" from [Beveridge and Joshi, 2011] and implement it into CCB pricing.

The regression is performed separately for S_t situations in different intervals, instead of a unified regression for all S_t situations , and we still use bases $\{S_t, {S_t}^2, F_t, {F_t}^2, Y_t, {Y_t}^2, S_tF_t, S_tY_t, F_tY_t\}$ for regression.

We divide S_t into the following four intervals for regression:

- $\Pi_1^j : (k_t, \infty)$,
- $\Pi_2^j : (C_t, k_t]$,
- $\Pi_3^j : (p_t, C_t]$,

- $\Pi_4^j : (-\infty, p_t]$,

where k_t is the call triggered price and p_t is the put triggered price.

Chapter 3

Empirical Studies on CCB Pricing

This chapter presents performance of the pricing algorithm by comparing the actual prices with the model prices of 10 CCBs with 3A credit rating in the first half of 2023. The experimental results confirm the effectiveness of "Multiple Regression" in Chapter 2. Then we use the predicted price as a determining factor for backtesting in the first half of 2023 and observe the rate of return.

3.1 Data Description

As of January 18, 2023, there are 487 CCBs circulating in the market and totally 30 AAA-rated CCBs. We select the top 10 3A rating CCBs with the highest trading volumes, covering 9 different major industries from manufacturing industry, transportation industry to banking sector. And we conducted daily pricing (using M=5000 paths for each pricing) from January 18, 2023, to July 17, 2023. The total number of pricing is 1180 for the 10 CCBs in 118 trading days.

In section 1, r is the average risk-free interest rate per day, we can easily get the

risk-free interest rate of an one-year government bond μ, then

$$r = (1+u)^{\frac{1}{252}} - 1.$$

In addition, σ is the so-called historical volatility of the stock over one year, calculated as follows: if we have historical time series of stock returns

$$r_t := \frac{S_t - S_{t-1}}{S_{t-1}}, t = 1, 2, 3 \dots T,$$

then the variance of these returns σ is defined as:

$$\sigma^2 = \frac{1}{T-1} \sum_{t=1}^{T} \left(r_t - \bar{r}\right)^2,$$

where

$$\bar{r} = \frac{1}{T} \sum_{t=1}^{T} r_t.$$

3.2 Performance of the Pricing Algorithm on 10 CCBs

We observed the "gap" between the model price and market price by MREs (Mean Relative Errors), MAREs (Mean Absolute Relative Errors) and RMSEs (Root Mean Square Errors), these three indicators are used to quantify the differences in pricing.

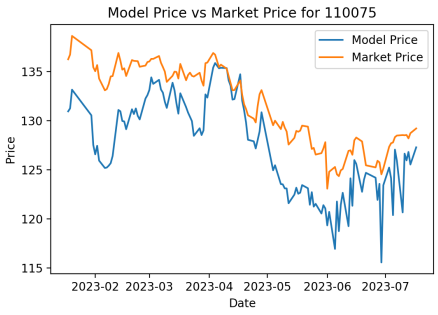
They are defined as follows:

$$\text{MRE} = \frac{1}{N} \sum_{i=1}^{N} \frac{V_i^{\text{mdl}} - V_i^{\text{mkt}}}{V_i^{\text{mkt}}},$$

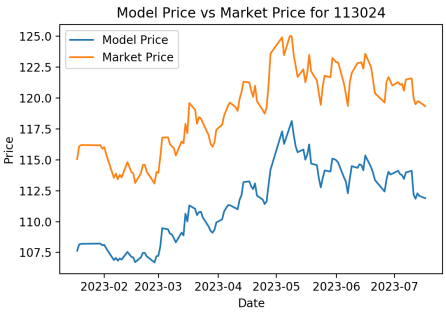
$$\text{MARE} = \frac{1}{N} \sum_{i=1}^{N} \frac{|V_i^{\text{mdl}} - V_i^{\text{mkt}}|}{V_i^{\text{mkt}}},$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\frac{V_i^{\text{mdl}} - V_i^{\text{mkt}}}{V_i^{\text{mkt}}}\right)^2},$$

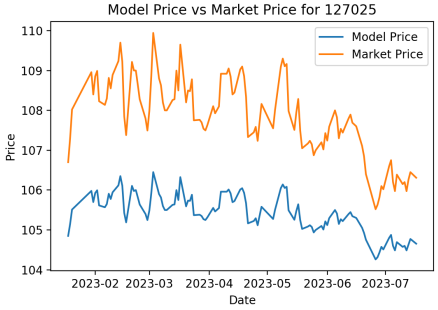
where V_i^{mdl} is the predicted CCB price on the ith trading day and V_i^{mkt} is the corresponding market price.



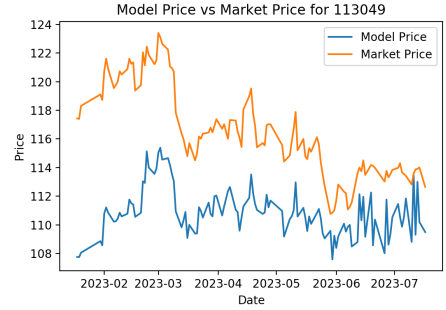
(a) Nan Hang



(b) He Jian

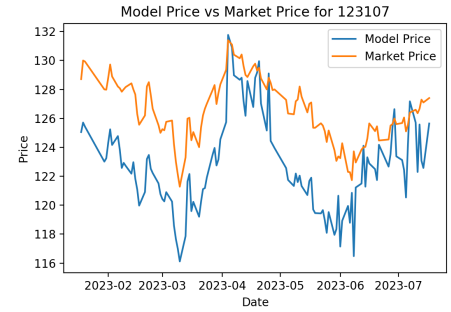


(c) Ji Dong

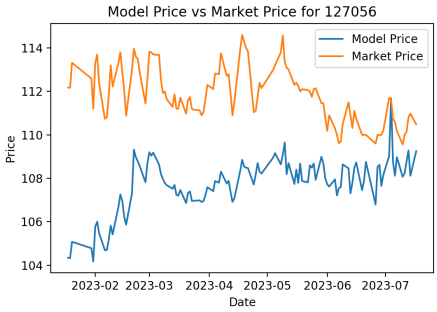


(d) Chang Qi

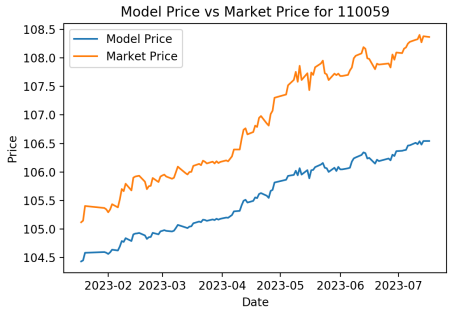
Figure 3.1: Model price vs market price for various CCBs. (Page 1)



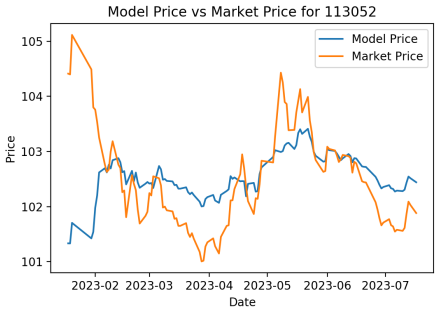
(a) Wen Shi



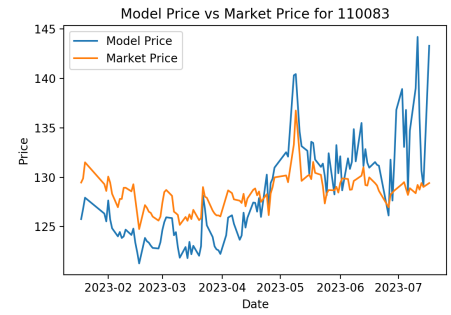
(b) Zhong Te



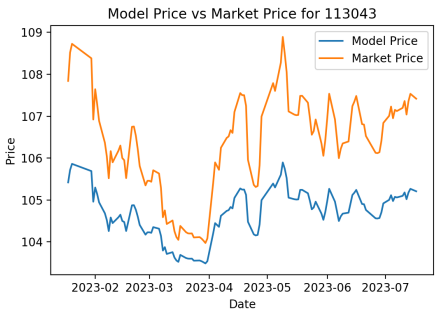
(c) Pu Fa



(d) Xing Ye



(e) Su Zu



(f) Cai Tong

Figure 3.2: Model price vs market price for various CCBs. (Page 2)

The average RMSE of 3.14 outperforms the result of 7.00 achieved by [Ma et al., 2020] using historical volatility.

If we use $\mathbf{e_t} = \{S_t, {S_t}^2, F_t, {F_t}^2, S_tF_t\}$ for regression rather than $\mathbf{e_t} = \{S_t, {S_t}^2, F_t, {F_t}^2, Y_t, {Y_t}^2, S_tF_t, S_tY_t, F_tY_t\}$, we can get the following result.

Table 3.1: The pricing errors for 10 different CCBs.

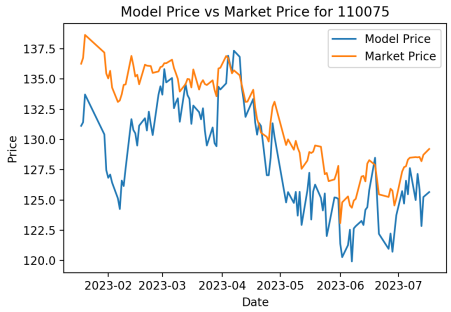
Bond Name	Ticker	MRE (%)	MARE(%)	RMSE(%)
Nan Hang	110075	-3.15	3.16	3.62
He Jian	113024	-6.24	6.24	6.26
Ji Dong	127025	-2.24	2.24	2.29
Chang Qi	113049	-4.78	4.78	5.18
Wen Shi	123107	-3.02	3.06	3.37
Zhong Te	127056	-3.67	3.67	3.95
Pu Fa	110059	-1.22	1.22	1.27
Xing Ye	113052	0.07	0.55	0.80
Su Zu	110083	-0.07	2.44	3.02
Cai Tong	113043	-1.57	1.57	1.66
Mean	-	-2.59	2.89	3.14

Table 3.2: The pricing errors for 10 different CCBs without base Y_t.

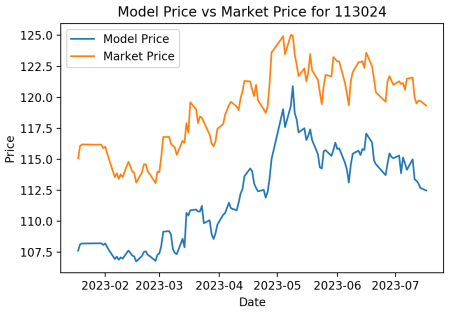
Bond Name	Ticker	MRE(%)	MARE(%)	RMSE (%)
Nan Hang	110075	-3.83	3.84	4.22
He Jian	113024	-6.41	6.41	6.43
Ji Dong	127025	-2.25	2.25	2.29
Chang Qi	113049	-5.87	5.87	6.07
Wen Shi	123107	-3.83	3.83	4.02
Zhong Te	127056	-4.49	4.49	4.61
Pu Fa	110059	-1.54	1.54	1.68
Xing Ye	113052	0.04	0.55	0.81
Su Zu	110083	-0.63	2.05	2.40
Cai Tong	113043	-1.57	1.57	1.66
mean	-	-3.04	3.24	3.42

After adding Y_t as the base, the average RMSE decreased from 3.42 to 3.14.

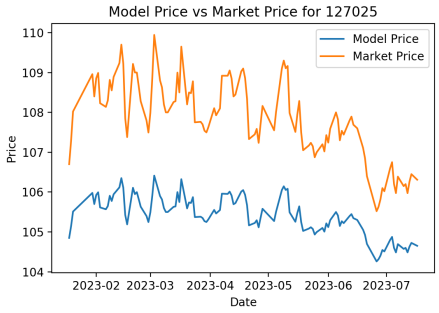
Now we redo the analysis with “Multiple Regression” using the same parameters and assumptions as before.



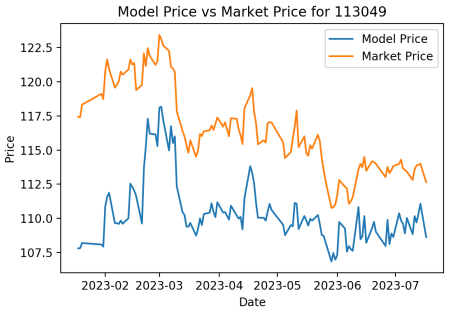
(a) Nan Hang



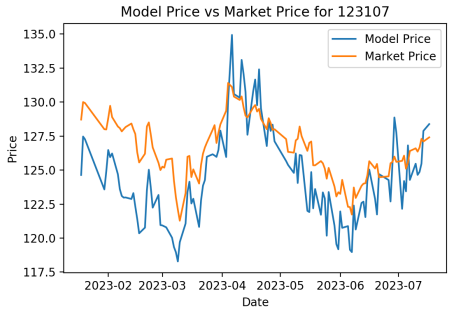
(b) He Jian



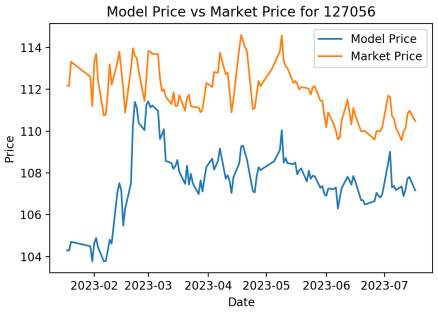
(c) Ji Dong



(d) Chang Qi

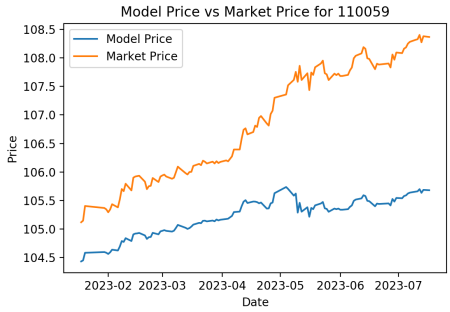


(e) Wen Shi

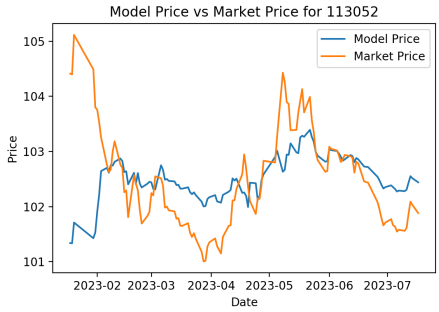


(f) Zhong Te

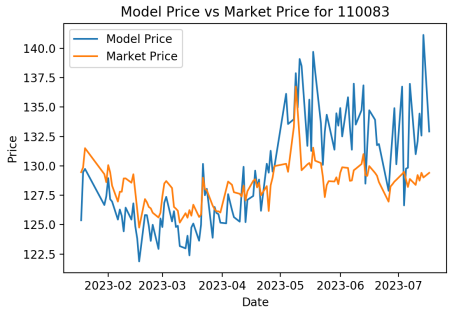
Figure 3.3: Model price vs market price for various CCBs. (Page 1)



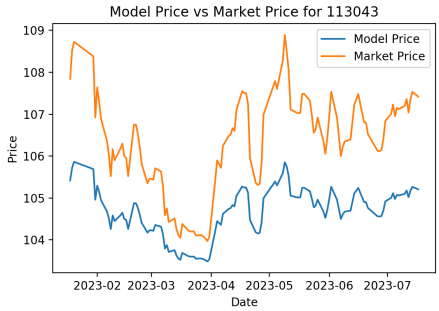
(a) Pu Fa



(b) Xing Ye



(c) Su Zu



(d) Cai Tong

Figure 3.4: Model price vs market price for various CCBs. (Page 2)

Table 3.3: The pricing errors for 10 different CCBs with "Multiple Regression".

Bond Name	Ticker	MRE(%)	MARE(%)	RMSE(%)
Nan Hang	110075	-2.45	2.50	2.93
He Jian	113024	-5.77	5.77	5.83
Ji Dong	127025	-2.25	2.25	2.29
Chang Qi	113049	-5.13	5.13	5.37
Wen Shi	123107	-1.66	2.02	2.33
Zhong Te	127056	-3.72	3.72	3.98
Pu Fa	110059	-1.52	1.52	1.66
Xing Ye	113052	0.04	0.56	0.82
Su Zu	110083	0.77	2.14	2.73
Cai Tong	113043	-1.57	1.57	1.66
mean	-	-2.32	2.72	2.96

From the graphs, it can be seen that the "gap" between model price and market price has narrowed, we can observe that the average RMSE decreases from 3.14 to

2.96 when “Multiple Regression” is applied, which indicates that it is useful in terms of practical pricing.

3.3 Trading Strategy by the Least Squares Factor

In this part, we use historical data to backtest and validate the predictive ability of our model price for CCBs. We use the model prices as a determining factor for backtesting and observe their returns : assume the transaction cost is 0.1%, long the most underpriced 10 CCBs among 487 CCBs in the market and rebalance daily from February 18, 2023 to July 17, 2023, totally 118 trade days.

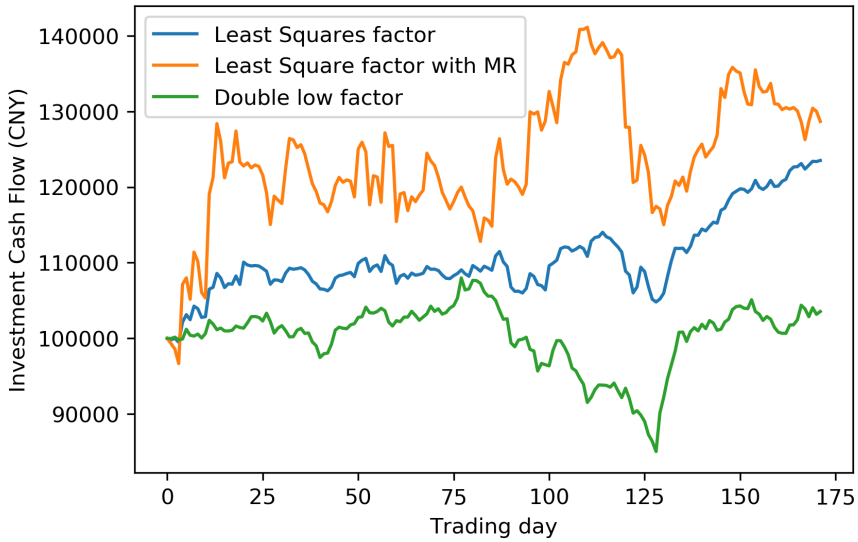


Figure 3.5: Least Squares factor backtesting for CCBs.

We select the commonly used “Double Low Strategy” as the baseline to observe the backtesting performance. It is evident that after implementing the “Multiple Regression” , the backtesting returns of Least Squares factor significantly improve.

	Cumulative Return(%)	Sharpe Ratio	Maximum Drawdown(%)
Least Squares	23.52	2.19	8.43
Least Squares with MR	28.68	1.10	20.45
Double Low	3.55	0.38	23.89

Table 3.4: Performance of Least Squares factors and Double Low factor.

Chapter 4

Multi-factor Strategy of CCBs

In this chapter, we aim to select factors that accurately reflect market information and propose a multi-factor quantitative model for CCB selection based on machine learning techniques. First, we conduct single factor IC analysis for daily frequency data and exemplify IC analysis cannot determine the monotonicity of factors, so we introduce a so-called "stratified backtesting" approach for single factor, i.e., for each given factor, allocate the same initial funds to n groups and each group holds all the CCBs of corresponding tier, then rebalance daily. Second, we use various multi-factor machine learning regression algorithms to predict CCB prices, select the algorithms with stable backtest returns, and finally use these algorithms to regress on factors that perform well in IC analysis and stratified backtesting to build our CCB multi-factor strategy.

Factor	Equation or Note
BP	Market price of the convertible bond
SP	Market price of the underlying stock
PR	Conversion premium rate, $PR = \frac{BP-CV}{CV}$
BR	Pure bond premium rate, $BR = \frac{BP-BV}{BV}$
D	Duration, $D = \frac{\sum_{t=1}^{n} tP_t/(1+i)^t}{\sum_{t=1}^{n} P_t/(1+i)^t}$, P_t is the coupon and/or principal payment, i is the market rate of return.
DL	Double low, $DL = BP + 100 \times PR$
YTM	Yield to Maturity
γ	Turnover rate
ϕ	Remaining size
σ	Stock volatility
β	Yesterday's rate of return
η	Last week's rate of return

Table 4.1: Factors referred to in Chapter 4.

4.1 Evaluation of Single Factor

IC refers to the correlation coefficient between the factor value and the rate of return in the subsequent trading day.

$$ic_t = cor(f_t, r_{t+1}), t = 1, 2, \cdots, n$$

where r_t is the rate of return at time t and f_t is the factor values at time t.

Then we do the following hypothesis testing,

$$\text{Null hypothesis } H_0: \quad ic_t = 0,$$
$$\text{Alternative hypothesis } H_1: \quad ic_t \neq 0.$$

IC values measures the linear relationship between the factor value and the rate of return in the next period, which reflects the predictive ability of the factor on returns in some way.

Information ratio (IR) is the ratio of the mean of IC values to its standard deviation, denoted as

$$ir = \frac{\bar{ic}}{\sigma_{ic}}.$$

where $\bar{ic}$ is the mean of ic and σ_{ic} is the standard deviation of ic.

IR values consider both the predictive ability of a factor on return (represented by the mean of IC) and the stability (represented by the standard deviation of IC).

Regarding the specific calculation of the correlation coefficient, we have two options: Pearson correlation coefficient and Spearman correlation coefficient.

First, Pearson correlation requires continuous data that follow a normal distribution while financial data often does not. Second, Pearson correlation is sensitive to outliers while Spearman correlation not. As a result, we only consider Spearman correlation coefficient.

In order not to create ambiguity, we call the correlation coefficient calculated using Pearson correlation coefficient IC, and the correlation coefficient calculated using Spearman correlation coefficient is called rank IC.

The pitfall of IC

The definition of IC suggests that a perfect CCB investor would have an IC = 1, while the most terrible picker might end up with an IC = -1.

It is well accepted that a realistic selection model can hardly have an IC materi-

ally different from 0 as a result of inherent difficulties of predicting Securities' return [Zhang et al., 2020]. Here, an IC value with an absolute value greater than 0.03 is considered significant.

Table 4.2: Example: IC analysis misjudgment.

$\mathbf{f}_t$	1.0	0.8	0.6	0.4	0.2	-0.2	-0.4	-0.6	-0.8	-1.0
$\mathbf{r}_{t+1}$	-0.8	-0.4	0.4	0.8	1.2	1.6	2.0	-1.2	-1.6	-2.0

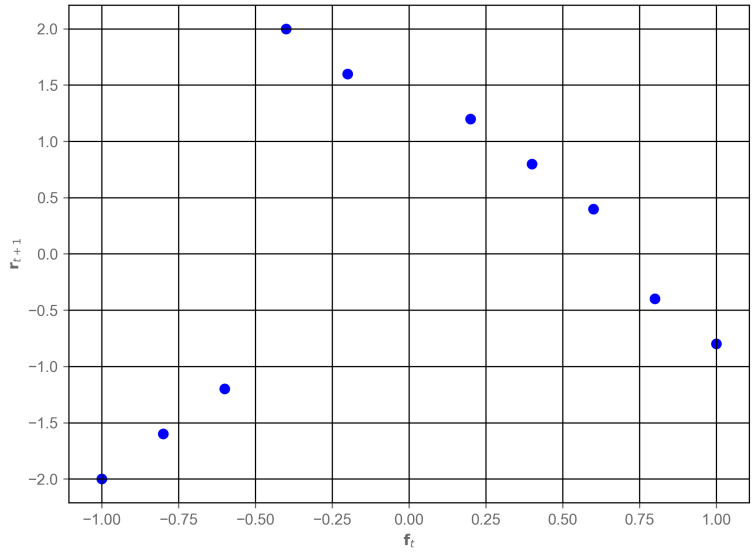
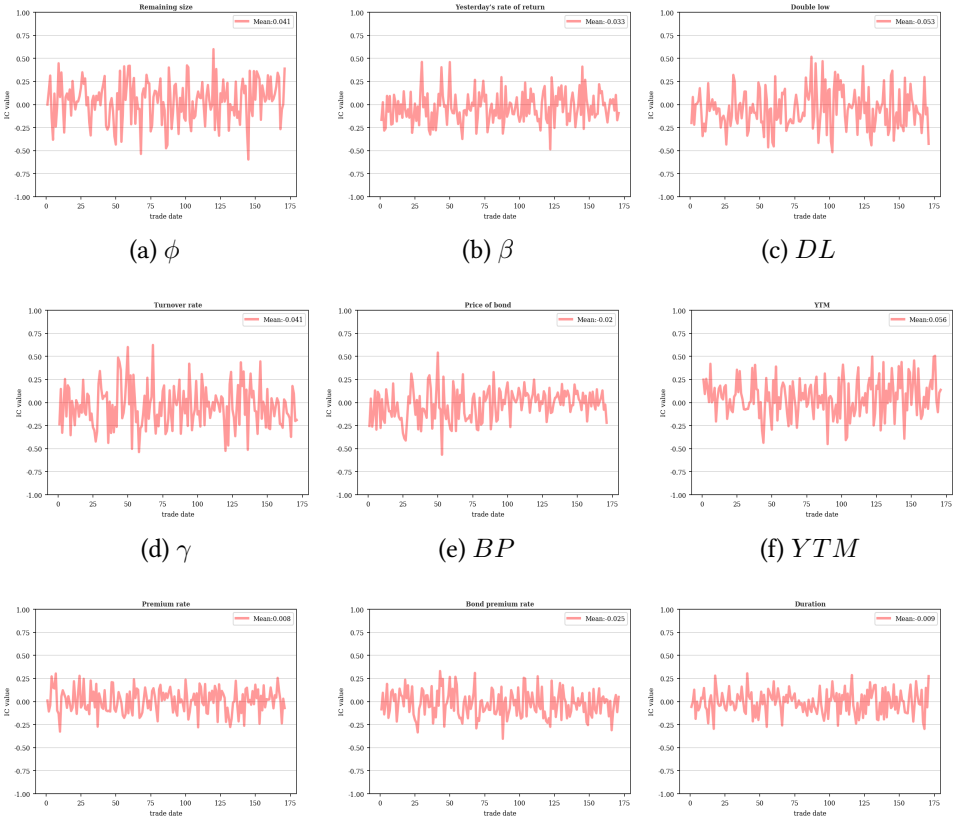


Figure 4.1: Scatter plot of this example.

In this example, IC value is 0.2909 while rank IC value is 0.3212. If we only consider the IC value as a single indicator, we would conclude that this factor has a strong CCB selection ability in the current period. However, when using the maximum factor value, it selected stocks with very poor returns. IC analysis fails to capture the monotonicity and stratification between factor values and returns. Therefore, we also need to do "stratified backtesting".

4.1.1 IC Analysis



(a) ϕ (b) β (c) DL

(d) γ (e) BP (f) YTM

(g) BR (h) PR (i) D

Figure 4.2: IC values with daily frequency.

In order to better observe the IC values, we take the mean value of the IC vector and record it as "IC" in the following table, and p value is the probability that H_0 holds true in the hypothesis testing on page 30.

factor	IC	IR	p value
β	0.031	-0.203	0.000
η	-0.034	-0.342	0.000
γ	-0.043	-0.183	0.000
ϕ	0.027	0.184	0.001
σ	-0.004	-0.089	0.611
BP	-0.002	-0.121	0.758
SP	-0.003	-0.032	0.699
PR	0.015	0.061	0.053
BR	-0.029	-0.176	0.000
DL	-0.051	-0.254	0.000
D	-0.018	-0.074	0.025
YTM	0.057	0.283	0.000

Table 4.3: IC mean and IR.

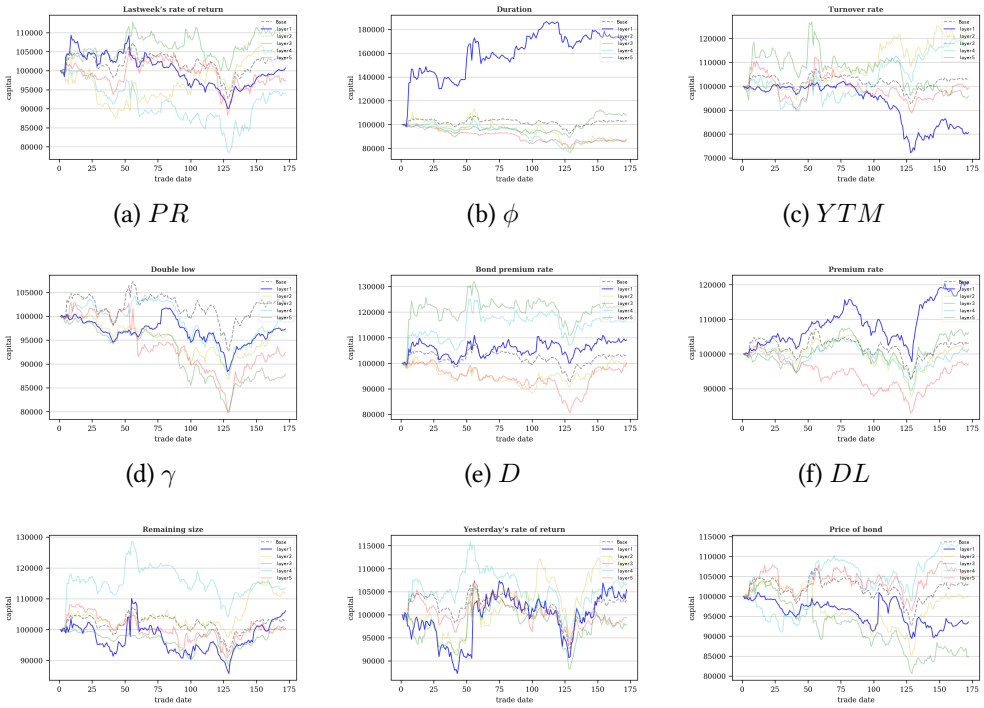
Generally speaking, the absolute value of IC mean value higher than 0.02 indicates a significant correlation. From the graphs above, it can be observed that factors such as β, η, γ, ϕ, DL , PR and YTM are highly correlated with CCB' returns.

4.1.2 Stratified Backtesting

All the CCBs are divided into 5 groups based on the values of the factor, and the backtest results are observed for each group.

The specific procedure is as follows:

- For each given factor, allocate the same initial capital to 5 groups for investment, and compound daily.
- Starting from August 10, 2020, divide the convertible bonds into 5 groups according to the factor value, and arrange the groups 1, 2, 3, 4, 5 in ascending order. Each group holds the corresponding group of convertible bonds and rebalance on the second day.
- Change the date to the next day until April 16th, 2021 , and repeat step 2.



(a) PR (b) ϕ (c) YTM

(d) γ (e) D (f) DL

(g) σ (h) BR (i) β

Figure 4.3: Stratified backtesting.

From the graphs, lower pure bond premium ratios are generally better and there is significant difference when the remaining size factor is minimal. Small-cap convertible bonds often exhibit active performance and possess growth potential [开源证券, 2021]. Higher YTM is generally better. Lower turnover rates are generally not preferred, as they indicate little to no returns. The relationship between duration and performance is not clear, but there is significant stratification, with the middle groups performing better. The double low factor shows a clear monotonic trend, and in general, lower values are better. There is no significant stratification in underlying stock volatility, This is consistent with the findings of [开源证券, 2021]. In theory, lower conversion premium rate is better, but the effect is not apparent. The optimal range for the conversion premium rate was 15% to 25% [西南证券, 2019], and also the effectiveness of the premium rate factor is subject to certain conditions [开源证券,

2021], so only relying on this factor will not lead to a good distinction. Yesterday's rate of return perform well, indicating that historical returns have predictive power for the future in the short term.

4.2 Multi-Factor Regression

4.2.1 Methodology

This part describes the framework of our approach and a collection of machine learning methods that we use in our analysis.

Multi-factor regression models:

$$R_{t,i} = f\left(x_{t-1,i}; \theta\right) + \epsilon_{t,i}$$

where $f(\cdot)$ defines a function with parameters , which represents the functional form in various machine learning and deep learning methods. $R_{t,i}$ represents the rate of returns of CCB i in period t, $x_{t-1,i} = (x_{t-1,i,1}, x_{t-1,i,2}, ..., x_{t-1,i,N})$ represents the factor vector of stock i in period $t-1$ and $\epsilon_{t,i}$ represents the error.

After determining a specific functional form $f(\cdot)$, a sliding window method is used to partition the training and testing datasets.

The specific procedure is as follows:

- Starting from August 10, 2020, train a machine learning model using data from the previous 7 days to obtain model parameters.
- Apply the trained model to the factor data on August 10, 2020, to predict the CCB's rate of return in the next day. Select the top N CCBs with the highest predicted returns for equal investment (investing the same amount of money).
- Change the date to the next day, and repeat steps 1 and 2.

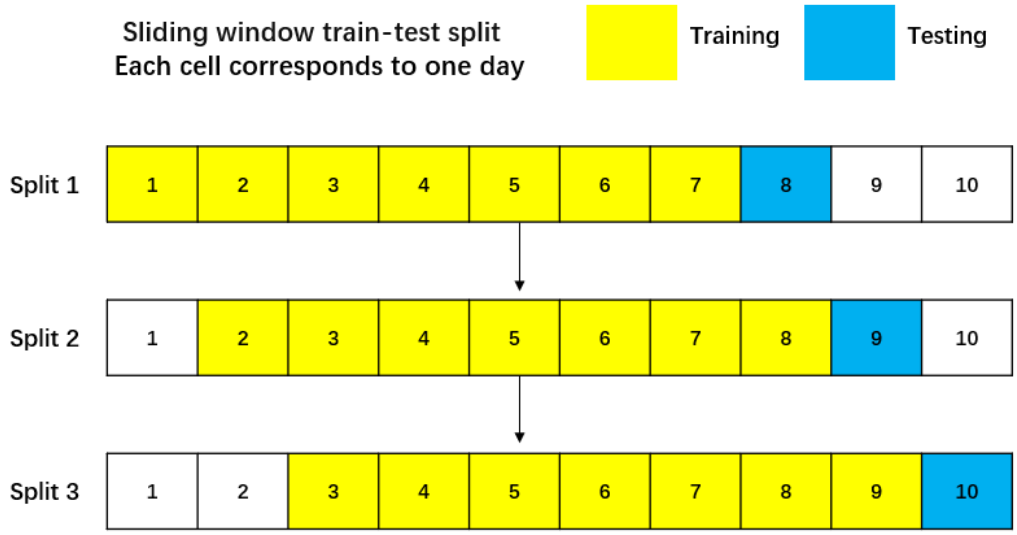


Figure 4.4: Sliding window train-test split of 7 days training set.

The default training set interval length is 7 days and we also examine the performance of models with training set lengths of 10 days and 20 days.

In terms of learning algorithms, we select linear learning models including linear regression, polynomial regression, as well as traditional machine learning algorithms such as SVM, random forests, XGBoost, and LightGBM, and they also include multi-layer perceptron (MLP). We aim to examine whether the performance improvement is positively correlated with the algorithm's complexity.

Although it is not possible to exhaust all machine learning algorithms, the selected representative algorithms have achieved good predictive performance in other domains. For example, MLP in computer vision [Tolstikhin et al., 2021], XGBoost in Control pile foundation design [Kavzoglu and Teke, 2022] and SVM in recognizing human actions [Schuldt et al., 2004].

The parameters selection of all machine learning model is done through grid search. Theoretically, as the window slides, the optimal parameters for each model will also change. Due to the high computational cost of grid search in each period,

we only performs parameter tuning in the training set of the first sliding window. In the subsequent window sliding process, the model parameters remain unchanged.

4.2.2 Data Source and Summary statistics

We select a sample of 400 CCBs listed on Shanghai and Shenzhen Stock Exchanges from August 3, 2020, to June 13, 2021, with daily frequency data.

The data is sourced from three APIs: `getMktConsBondPerf`, `getBondConvDer`, and `getBondRating` provided by Tonglian Data. Twelve fundamental factors are selected as the input variables. By combining the factor features of each convertible bond at period $t-1$ with the rate of return label at period t, a complete dataset is created.

After removing convertible bonds with missing factor values and outliers by 3 σ principle, there are 13017 valid samples remaining after handling missing values. Due to the small number of data features, the corresponding convertible bonds were directly removed instead of missing factor being filled with zeros.

The following is the basic statistical information of the data.

Feature	Count	Mean	Std	Min	25%	50%	75%	Max
Price of bond	13017	112.39	17.54	71.68	100.79	108.74	119.13	208.69
Yesterday's rate of return	13017	-0.08	1.47	-8.78	-0.65	-0.04	0.46	8.89
Lastweek's rate of return	13017	-0.44	3.17	-15.88	-1.84	-0.25	1.01	15.32
Premium rate	13017	25.31	22.72	-3.14	10.15	20.46	32.50	169.61
Bond premium rate	13017	22.26	23.03	-24.34	6.54	17.00	32.79	148.66
Dowble low	13017	137.70	20.58	74.61	125.32	133.28	143.99	245.45
Duration	13017	4.09	0.76	1.96	3.46	4.21	4.72	5.36
Turnover rate	13017	10.04	29.17	0.00	0.45	1.10	4.16	275.83
Remaining size	13017	8.45	7.82	0.53	2.99	6.03	9.60	31.50
YTM	13017	0.88	3.83	-12.34	-0.94	1.30	3.39	12.96
Stock volatility	13017	46.18	28.02	0.00	25.44	40.76	65.39	143.45
Price of stock	13017	10.74	7.19	1.63	6.01	8.47	12.66	39.17

Table 4.4: Summary statistics. "Count" represents the total number of valid entries, "Mean" represents the average value, "Std" represents the standard deviation, "Min" represents the minimum value, "25%", "50%", and "75%" represent the corresponding quartiles, and "Max" represents the maximum value.

4.3 Numerical Experiments

Before doing machine learning, we first consider the factor correlation.

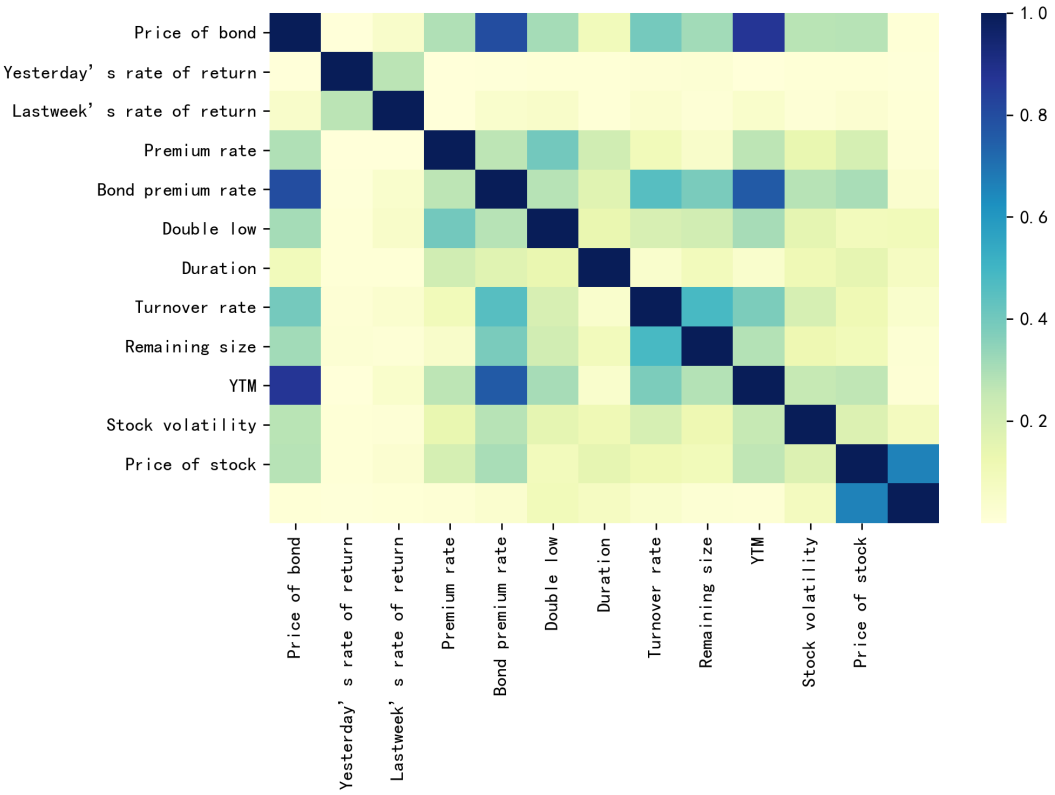


Figure 4.5: Factor correlation.

"Price of bond" has a high correlation with both "YTM" and "Bond premium rate", so they should not be selected as factors together. Here, we disregard factor "Price of bond", because its IC value is relatively small.

We apply the strategy to simulated trading from August 3, 2020, to April 16, 2021, totaling 172 trade days with a 0.1% transaction cost. The x-axis represents the number of days (starting from August 3, 2020), and the y-axis represents our investment capital (initial capital is 100, 000), and we get the "Market Index" by investing all the CCBs in the data with equal capital, and rebalance daily.

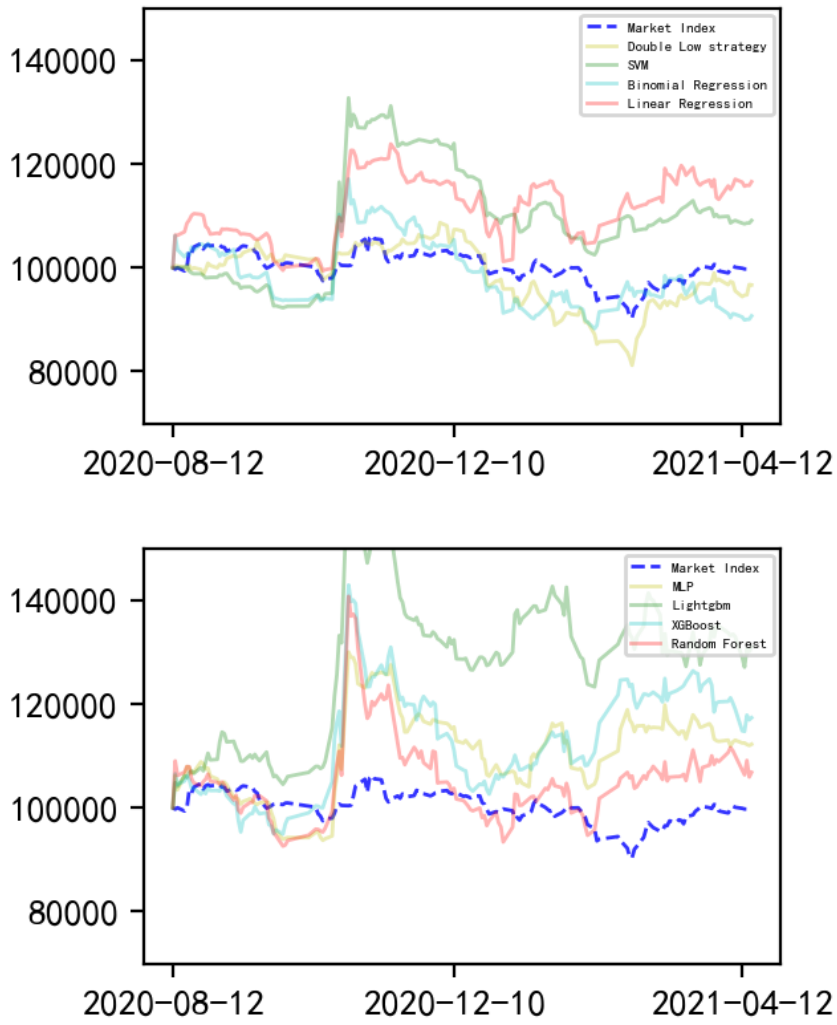


Figure 4.6: Results of multi-factor machine learning backtesting for 7 days training set.

Table 4.5: Backtest metrics for 7 days training set.

Algorithm	Annualized Return (%)	Sharpe Ratio	Maximum Drawdown(%)	VAR(%
Market Index	0.02	-4.59	14.94	1.42
Double Low	3.53	-1.18	25.29	2.00
SVM	14.18	2.31	22.79	1.87

Binomial Regression	-13.86	-7.36	24.62	2.58
Linear Regression	26.34	6.67	18.28	2.02
MLP	-35.24	-16.77	33.48	2.86
Lightgbm	51.06	9.28	27.64	2.68
XGBoost	27.63	4.64	28.18	3.43
Random Forest	10.57	0.91	33.63	3.04

VAR(%) is the value at risk for $\alpha = 5\%$.

Performance when adjusting the width of the sliding window

In addition to 7 days training set , we also compare the results with 10 days and 20 days training set and examine the backtesting metrics for different time lengths of training data.

Table 4.6: Backtest metrics for 10 days training set.

Algorithm	Annualized Return (%)	Sharpe Ratio	Maximum Drawdown(%)	VAR(%)
Market Index	0.01	-0.73	13.53	1.38
Double Low	2.30	1.57	24.01	1.93
SVM	16.31	3.08	22.78	2.20
Binomial Regression	-2.68	-2.12	28.75	2.95
Linear Regression	59.42	16.74	15.09	2.17
MLP	-3.88	-6.34	12.59	1.48
Lightgbm	-6.48	-5.08	16.64	1.38
XGBoost	0.63	-2.09	16.03	1.59
Random Forest	-11.43	-7.19	20.44	1.67

Table 4.7: Backtest metrics for 20 days training set.

Algorithm	Annualized Return (%)	Sharpe Ratio	Maximum Drawdown(%)	VAR(%)

Market Index	0.01	-4.59	14.94	1.42
Double Low	2.30	-1.18	25.29	2.00
SVM	20.93	7.26	13.98	1.54
Binomial Regression	-24.85	-12.66	32.29	2.77
Linear Regression	45.20	12.52	22.23	2.35
MLP	23.55	6.52	12.28	2.67
Lightgbm	15.67	2.73	29.53	2.67
XGBoost	49.06	10.81	17.60	2.63
Random Forest	36.73	8.78	19.74	2.49

It can be observed that as the training set size increases, the profitability of non-linear methods gradually improves. However, the performance of SVM and linear regression remains consistently good. To save computational time, when developing our own strategy in the next section, we choose 7 days training set and SVM and linear regression as our regression algorithms.

4.3.1 Validation of Effectiveness

In this part, we do validation of effectiveness of the above multi-factor model.

Based on the previous conclusions, we build our own multi-factor strategies, construct composite factors and observe the backtesting results, and then apply the factors to data in the first half of 2023 to observe the returns.

We uses SVM and linear regression to regress the factors “Conversion Premium Rate”, “Double Low ”, “Bond Premium Rate”, “YTM” , and “Remaining Size” to obtain a predicted price, and then long the most underpriced 10 CCBs by underpricing ratio among CCBs in the data and rebalance daily.

The specific procedure is as follows:

- Starting from February 6, 2023, train the model using data from the previous 7 days to obtain model parameters.
- Apply the trained model to the factor data on the next day, to predict the stock returns.
- Select the top 10 convertible bonds with the highest predicted returns and their "Double Low" values are below 180 (through extensive testing, we have found that this method can significantly improve the return on investment), then investing the same amount of money in each convertible bond. And rebalance on the second day.
- Change the date to the next day, and repeat steps 1, 2 and 3.

Assuming the transaction cost is 0.1%, we apply the strategy to simulated trading from February, 2023 to July 17, 2023, totally 118 trading days, and get the market index by invest all the convertible bonds in the data with equal capital, and rebalance daily.

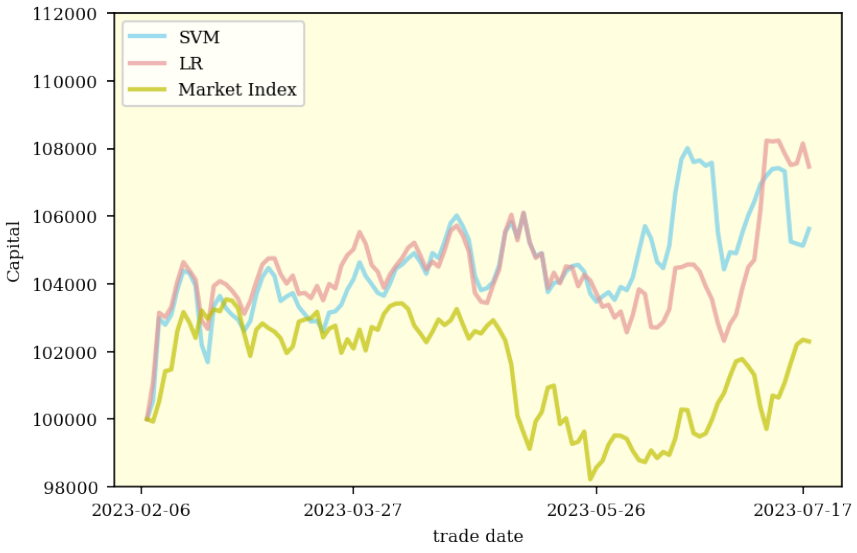


Figure 4.7: Simulation of trading for SVM and linear regression.

Algorithm	Annualized Return (%)	Sharpe Ratio	Max Drawdown (%)	VAR (%)
SVM	13.22	3.06	3.32	0.94
LR	18.62	6.57	3.57	0.71

Table 4.8: Backtest metrics of two multi-factor strategies.

Chapter 5

Conclusion

The study contributes to the literature of CCB pricing. In CCB research, the downward adjustment clause is often the most difficult to consider. Taking practical situations into account, in order to avoid financial distress upon put provision, bond issuers can use the downward adjustment clause to lower the conversion price. Therefore, we treat the downward adjustment clause as a probabilistic event triggering the put provision. In this way, we combine the downward adjustment clause with put provision in a simple manner.

Also, we confirmed the practical effect of the machine learning-driven convertible bond factor strategy. Different from most Chinese brokerages and literature practices, we conducted systematic empirical analysis of the factors in CCBs, identifying factors that truly have potential profit effects. We used various machine learning methods to predict CCB prices and construct our own strategies.

We need to acknowledge the limitations of our thesis, the insufficient amount of data and the tendency for overfitting when using XGBoost and MLP. In the subsequent work, we will increase the amount of data. However, this process is theoretically simple because our framework has been well constructed, requiring only a change in the input data and parameter modification.

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